

Running a Sample Workload

Running a Sample Workload with Docker

After you install and configure the toolkit and install an NVIDIA GPU Driver, you can verify your installation by running a sample workload.

- Run a sample CUDA container:

```
sudo docker run --rm --runtime=nvidia --gpus all ubuntu nvidia-smi
```

Your output should resemble the following:

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												
NVIDIA-SMI 535.86.10				Driver Version: 535.86.10				CUDA Version: 12.2				
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												
GPU	Name	Persistence-M		Bus-Id		Disp.A		Volatile	Uncorr.	ECC		
Fan	Temp	Perf	Pwr:Usage/Cap		Memory-Usage		GPU-Util		Compute M.			
										MIG M.		
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												
0	Tesla T4	On		00000000:00:1E.0		Off				0		
N/A	34C	P8	9W / 70W		0MiB / 15109MiB		0%		Default			
										N/A		
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												
Processes:												
GPU	GI	CI	PID		Type	Process name				GPU	Memory	
		ID	ID							Usage		
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												
No running processes found												
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												

Running a Sample Workload with Podman

After you install and configure the toolkit (including [generating a CDI specification](#)) and install an NVIDIA GPU Driver, you can verify your installation by running a sample workload.

- Run a sample CUDA container:

```
podman run --rm --security-opt=label=disable \
  --device=nvidia.com/gpu=all \
  ubuntu nvidia-smi
```

Your output should resemble the following:

```
+-----+
| NVIDIA-SMI 535.86.10      Driver Version: 535.86.10      CUDA Version: 12.2      |
+-----+-----+-----+-----+-----+-----+
| GPU  Name      Persistence-M| Bus-Id      Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf  Pwr:Usage/Cap|      Memory-Usage | GPU-Util  Compute M. |
|                               |                   |              MIG M. |
+-----+-----+-----+-----+-----+-----+
|    0   Tesla T4          On   | 00000000:00:1E.0 Off |              0      |
| N/A   34C    P8          9W / 70W |      0MiB / 15109MiB |      0%      Default |
|                               |                   |              N/A   |
+-----+-----+-----+-----+-----+-----+

+-----+
| Processes:                                     |
|  GPU   GI    CI          PID    Type    Process name                        GPU Memory |
|          ID    ID                                   |          Usage |
+-----+-----+-----+-----+-----+-----+
|  No running processes found                                     |
+-----+
```

Running Sample Workloads with containerd or CRI-O

These runtimes are more common with Kubernetes than desktop computing. Refer to [About the NVIDIA GPU Operator](#) in the NVIDIA GPU Operator documentation for more information.